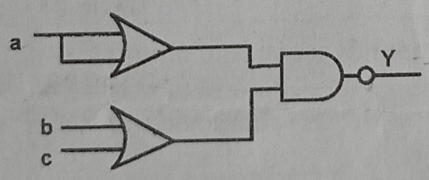
Total Printed Pages: 03

CSA-102
Examination –Dec- 2023
B.Tech. I : CS, IT, AI, BC, IoT
Digital Electronics

Time : 2 Hrs

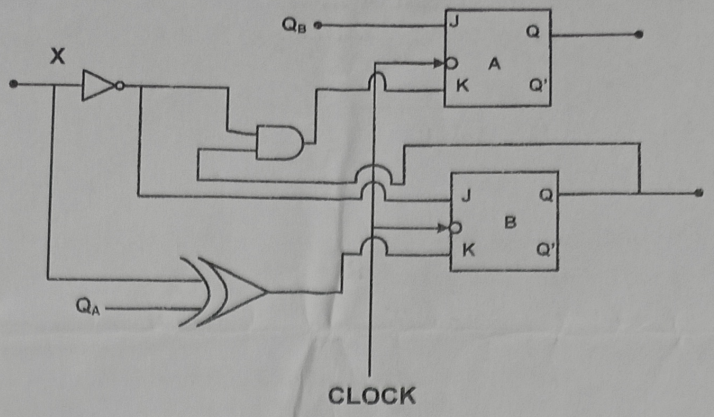
Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	(i) The 2's complement of the no. 1011010 is..... (ii) Convert $(11011)_{gray}$ to $(N)_{binary}$	02	1,2	1
	(b)	(i) Represent 6 in 8 4 -2 -1 code (ii) Convert the $(545)_D$ in BCD	02	1,2	1
	(c)	BCD addition - $(55)_{10} + (74)_{10}$	03	3	1
		OR			
		Drive all gates from NAND gate, draw the truth table also.	03	3	1
	(d)	Convert the following: (a)Convert $(21FA.3A)_{16}$ to $(N)_8$ (b) Add $(756.23)_8 + (ABC.AA)_{16}$ (c) Subtract $(011011)_2 - (110011)_2$ (d) $(1721.54)_8$ to $(N)_{10}$ (e) $(147.35)_{10}$ to $(N)_2$	05	4	1
		OR			
		A 7 bit hamming code 1110101 is received. Is there is any error. If yes locate the position. What is the actual data send. Considered the even parity.	05	4	1
Q.2	(a)	(i) The output Y is  (ii) Apply demorgan's theorem- $\overline{(A+B)\,\overline{C}\,\overline{D}+\overline{E}+F}$	02	1,2	2

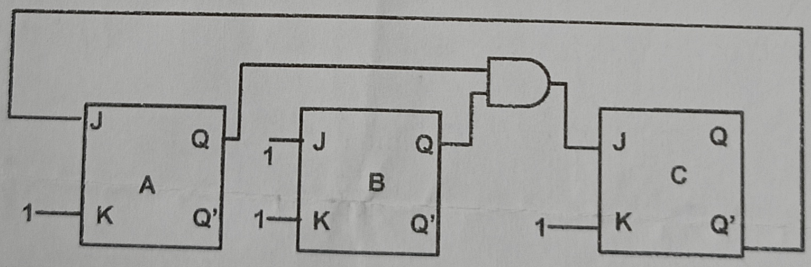
	(b)	The simplified expression consist of ..... no. of octave, ... square, ...no. of pair, ...no. of single. $F = \Sigma(0,2,4,5,6)$			
	(c)	If $F = A'B'C' + AB'D + A'B'CD'$ $d = ABC + AB'D'$ find POS.	03	3,4	2
		OR		3,4	
		If $F = \pi(1,3,5,7,13,15)$, find SOP.	03	3,4	2
	(d)	If $F = A'B'C' + AB'D + A'B'CD'$ $d = ABC + AB'D'$ Implement it with the NOR Gate	05	3,4	2
		OR			
		If $F(A, B, C, D) = \Sigma m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d(0, 2, 14)$ find the SOP & POS. Also draw the logic diagram for both the output.	05	3,4	2
Q.3	(a)	(i) For a design of 16x1 multiplexer, no. of select line required..... (ii) The given no. 101101 has which parity?	02	1,2	3
	(b)	What is encoder?	02	1,2	3
	(c)	Design a full adder with two half adder. Draw truth table also.	03	4	3
		OR			
		Construct a 5x32 decoder using four 3x8 decoder and one 2x4 decoder.	03	4	3
	(d)	Design a full substractor using multiplexer.	05	5	3
		OR			
		Design a three bit binary to gray code converter.	05	5	3
Q.4	(a)	(i) If an active-HIGH S-R Flip Flop has a 1 on the S input and a 0 on the R input then the output is ..... (ii) Name the logic circuit (flip flop) does not have no change condition.	02	1,2	4
	(b)	What is the difference between combinational & sequential circuit?	02	1,2	4
	(c)	Explain the working of JK flip flop, with its characteristic table and excitation table.	03	3,4	4
		OR			
		Convert SR to T flip-flop.	03	4	4

(d) Draw the state diagram



OR

		Explain different type of shift register explain them with neat diagram.	05	5	4
Q.5	(a)	(i) A decade counter require .....no. of flip flop. (ii) A .....is a cascade arrangement of flip flop where the output of one flip flop drives the input of the other flip flop.	02	1,2	5
	(b)	What is the difference between synchronous and asynchronous counter?	02	1,2	5
	(c)	Design a 3 bit asynchronous up counter.	03	5	5
		OR			
		Design a synchronous counter 0,3,5,6.	03	5	5
	(d)	Design a synchronous mode 6 up counter.	05	5	5
		OR			
		Find the mode of the counter and check whether it is locked out or not. Initial state is 000.	05	5	5



Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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